Life Without Driving

Everyday access without a car



Can I finish my outing—and get home the same day?

Japan ODPT Challenge 2026: A competition to build services using public transport open data to address local mobility problems

29.4%
of Japan is aged 65+

51.5%
of municipalities designated depopulated

~20%
of regional-area households have no car

Who feels the problem?

Residents without reliable access to a car

Older people who stop driving; other non-drivers.
A late appointment can mean a missed return.

Visitors and cross-boundary travellers

Arrive without a car; reach a hospital, school or visit beyond the city line and return.

The municipal transport officer

Needs evidence on where service fails and what a change helps.

Existing tools cover only parts of the task

Tool	Capabilites and Limitations
Google Maps	plans transit routes but requires separate outward and return checks.
Komibasu	supports bus-route design; complete outing checks remain unverified.
Traffic Echo	maps service gaps by time and frequency; complete outing checks remain unverified.
LINKS Mobilys	evaluates service scenarios; activity-and-return checks and failure explanations remain unverified.
Shiwaku 250 m map	shows walking access, but excludes timetables and on-demand transport

Our proposed contribution: activity + return conditions, explicit failure reasons, and the effects of a service change.

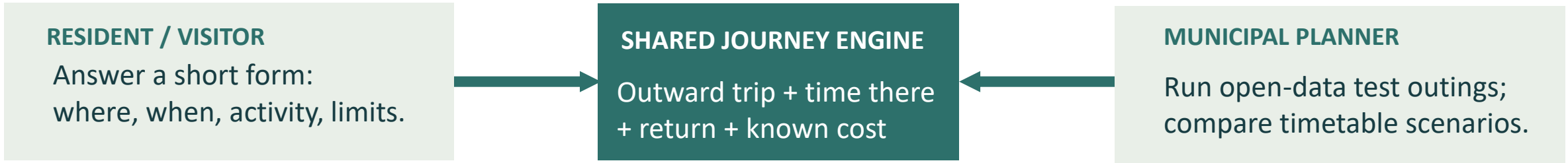
One outing test, two useful workspaces

A responsive web app for a resident’s phone and a transport officer’s desktop.

TO Enable residents and visitors without a car to complete activities and return the same day within their time and budget limits

BY Showing the municipality where and why modelled outings fail under actual service conditions, and how many more would complete if a condition changed.

USING An application built on public transport open data, a shared journey engine, and evidence-based explanations.



User journeys and the shared app flow

Two flows, one calculation engine. Read each column from top to bottom. The resident and planner can use the app independently.

Before deployment: import open data and precompute the planner baseline (steps 1–3) via background batch jobs; residents do not need to supply it..

DATA IN → Annaka bus GTFS + operator rules → versioned SQLite. ODPT adds Anna-car GTFS-Flex + ride logs.

	1 DESCRIBE	2 CONFIRM / RUN	3 UNDERSTAND	4 DECIDE / REPEAT
USER ACTIONS	Resident: fill outing form. Planner: select test cases and a scenario.	Confirm date + visit time, walking, return deadline and cost limit.	Review result + reason. Ask AI to explain it. Read the cited evidence.	Resident: revise a limit. Planner: inspect outings helped or harmed.
INSIDE THE APP	**Web UI → FastAPI** Validate fields and role. Create a single request or a modelled test batch.	**Python + pandas** Read SQLite schedules. Check out + visit + back. Test a copied timetable.	**RAG + LLM** Retrieve service rules. Explain the code result. Cite or state unknown.	**SQLite + batch jobs** Version test results. Recompute the map. Protect private records.
	L3 L6 L7 L11	L4 L5 L8	L6 L10 L12	L5 L8 L14

Bounded first pilot

Model selected outings before claiming town-wide coverage.

SERVICE AREA + DATA

ANNAKA MUNICIPALITY, GUNMA PREFECTURE. 52,578 residents; 37.6% aged 65+

Selected origin areas / stops → 3–5 destinations hospital, shop, civic

An-bus: fixed-route GTFS + official rules.
Anna-car: on-demand zones, hours and booking rules.

20–30 DOCUMENTED CASES

Each has a date, origin, destination, visit duration, return/walking limits, manually checked result + sources.

WHAT FITS BY THE CLASS FINAL

1.125 core - data pipeline, journey engine, secure workspaces and AI evaluation.

Class stretch / ODPT core: on-demand eligibility and journey checks, subject to booking availability.

2–3 PREDEFINED TIMETABLE CHANGES

Re-run the same cases on baseline and copied schedules.

Main uncertainty / risk	Response
Missing or stale data	Audit sources and dates. Flag gaps and uncertainty.
Round-trip routing	Verify full outings first. Limit transfers if needed.
Unvalidated user needs	Use proxy testing. Avoid population-level claims.

A five-week build with visible proof

Targets below are proposed acceptance checks—not results already achieved.

MILESTONES

BY 8 NOV

Audit feed + five outings.
Freeze input fields and sources.

~19 NOV · MIDPOINT

Working engine + both workspaces.
Role checks + baseline map.

BY 25 NOV

Freeze cases and scenarios.
Run AI tests; fix critical errors.

~3 DEC · FINAL

Held-out checks + repeatable demo.
Setup guide and limits.

WHAT THE FINAL 1.125 DEMO WILL SHOW

Check	Proposed pass condition
Journey correctness	20-30 manually checked cases agree, including a held-out subset.
Scenario integrity	Repeatable before-and-after results include journeys made worse.
AI reliability	20 test prompts: zero invented critical facts, with citations or explicit uncertainty
Access control	Direct UI, API and AI-tool tests block other users' private data and unauthorised scenario changes.
Usability + repeatability	3 proxy testers complete the core tasks; another student can run the documented demo from the same data snapshot.

ODPT NEXT · December–January: integrate Anna-car + ride logs, evaluate duration estimates, expand coverage and polish the public demo.

END